Multimodal Livestock Insurance Framework Using Muzzle Biometrics, Disease Features and Clinical Indicators

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Abstract

This study presents a multimodal framework designed to support livestock insurance assessment by integrating clinical attributes, behavioural patterns and image-based biometric features. The system is built around three specialised models, each one tailored to the nature of the data it receives. A convolutional network trained on cow–muzzle images achieved highly stable convergence through a combination of dataset restructuring, strong augmentation and focal-loss training. A second model, trained on a cattle-disease dataset, learned visible health indicators across multiple classes. A Random Forest classifier, calibrated using Platt scaling, generated smooth probability estimates for structured clinical values. The outputs of these three branches were combined to form a unified risk indicator that captures different aspects of the animal's condition.

The results show that each branch contributes a consistent and interpretable component of the final assessment. The visual model displays strong biometric reliability, the disease model captures symptom-based differences and the structured-data model supplies calibrated numerical estimates. Together, the system demonstrates that multimodal intelligence offers a more complete, transparent and practical foundation for livestock insurance workflows. The work establishes a clear direction for future extensions, including real-time monitoring, additional sensor inputs and automated claim processing.

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0.1 INTRODUCTION

Livestock contributes or helps significantly to rural income, food security, and the national agricultural economy. Farmers and insurance providers face recurring financial risks due to disease outbreaks, accidents, and unanticipated mortality events. Conventional livestock insurance procedures rely on manual inspection and written documentation, creating delays in verification and leaving considerable room for inconsistencies in claim assessment. These limitations often result in slow settlement cycles and increased vulnerability to fraudulent claims.

Recent developments in artificial intelligence and sensor-based livestock monitoring have opened new possibilities for improving the reliability of insurance processes. Biometric identification techniques, including muzzle pattern analysis and facial imaging, have shown strong potential for establishing individual animal identity with high precision [1, 7]. Surveys on livestock biometrics report that muzzle textures, gait signatures, and depth-based imaging can uniquely distinguish one animal from another, enabling accurate and continuous identity tracking [4, 5]. At the same time, advances in Precision Livestock Farming technologies provide access to temperature readings, feeding activity, and behavioural indicators that reflect real-time health status [2, 3].

Integrating these modalities creates a data-rich environment suitable for risk modelling in insurance. Machine learning enables the interpretation of disease indicators, feeding deviations, abnormal movements, and thermal patterns. When combined with biometric identity verification, these signals allow insurers to evaluate claims objectively and assign premiums that reflect the condition of individual animals rather than broad breed-level assumptions. Blockchain platforms further enhance reliability by maintaining transparent and tamper-resistant records of health events, ownership, and historical claims [6].

The purpose of this study is to develop a multimodal framework that bridges biometric recognition, behavioural sensing, and machine learning-based decision support for livestock insurance. Three datasets—disease attributes, feeding and activity logs, and muzzle images—are employed to demonstrate how complementary information streams can be fused for identity verification and health-risk assessment. The system presented here aims to strengthen transparency, reduce fraudulent activity, and provide insurers with a consistent basis for personalised coverage decisions. The resulting approach aligns with current trends in livestock technology research while offering a practical pathway for modernising insurance operations.

0.2 RELATED WORK

Research on livestock insurance and animal identification has progressed rapidly over the last decade, driven by advances in computer vision, biometric analysis, and sensor-based livestock monitoring. Several studies have highlighted the need for reliable animal identity verification, particularly in insurance and traceability applications. Ahmad et al. [1] demonstrated that biometric identification can replace manual verification processes, reducing the dependence on subjective inspection and signif-

icantly lowering the probability of fraudulent claims. Their work established the importance of linking each insured animal to a unique and verifiable digital identity.

Biometric research in livestock has expanded to multiple modalities, including muzzle patterns, facial images, and gait signatures. A comprehensive survey by Meng et al. [4] examined computer vision techniques for distinguishing individual cattle and found that muzzle pattern analysis remains one of the most stable identifiers across age and environmental conditions. Shojaeipour et al. [7] developed automated muzzle detection and presented a practical biometric pipeline suitable for real farm conditions, which supports its relevance for insurance verification.

Depth-data based biometric models have also gained attention. Sharma et al. [5] introduced a metric-learning approach that uses three-dimensional depth features to improve robustness under variable lighting. These findings indicate that combining two-dimensional and three-dimensional cues strengthens identity verification, particularly for field conditions where image quality is inconsistent.

Alongside computer vision, Precision Livestock Farming (PLF) technologies provide continuous streams of behavioural and physiological data. Reviews by Aquilani et al. [2] describe the integration of temperature sensors, feeding monitors, accelerometers, and environmental devices to capture real-time changes in health. Similarly, Curti et al. [3] outlined how wearable sensors and machine learning enable early disease detection, offering benefits for risk assessment and welfare management.

Blockchain and distributed ledger technologies have also been explored as mechanisms for strengthening transparency in livestock insurance. Shen et al. [6] demonstrated that combining IoT sensors with blockchain records supports reliable claim validation and ensures that health events cannot be altered retroactively. Their findings are particularly relevant for building tamper-resistant insurance systems.

The current body of literature shows progress in biometric identification, health monitoring, behavioural modelling, and transparent data management. However, existing studies often examine these components independently. There remains limited research that integrates biometric identity recognition, sensor-driven health analysis, and machine learning based risk modelling into a single end-to-end framework for livestock insurance. The present work seeks to address this gap by developing a multimodal system that unifies these components into a cohesive insurance-oriented pipeline.

0.3 DATA SOURCES FOR HEALTH-DRIVEN INSURANCE MODELLING

Accurate livestock insurance decisions require a detailed understanding of an animal's health condition, daily behaviour, and vulnerability to disease. Traditional insurance processes rarely have access to such information and rely instead on periodic inspections and general breed-level statistics. To enable individualised risk evaluation, this study uses two complementary data sources that capture clinical health indicators and long-term behavioural patterns. Together, these datasets provide the information necessary to estimate health risk in a systematic and data-driven manner.

0.3.1 DISEASE AND CLINICAL ATTRIBUTES DATASET

The first dataset contains structured information describing various clinical and environmental factors associated with cattle diseases. Each entry includes attributes such as breed, age, visible symptoms, production stage, recent feeding disturbances, and external stressors. These records allow the model to differentiate between mild conditions, high-risk symptoms, and patterns that typically precede critical health events.

Since the dataset includes both numerical and categorical variables, it is well suited for ensemble learning techniques such as Random Forests. These models can interpret mixed-type features and capture non-linear relationships between symptoms and disease outcomes. The output of this stage is a disease-risk score for each animal. This score reflects the likelihood of disease under the current conditions and provides a structured basis for premium adjustment in an insurance context.

0.3.2 BEHAVIOURAL AND FEEDING PATTERN DATASET

The second dataset focuses on continuous observations collected over multiple days. It includes measurements of feeding quantity, water intake, body-temperature variations, activity cycles, and other behavioural indicators. Fluctuations in these patterns are often the earliest signs of illness, appearing before external symptoms become visible.

Because the data form time-dependent sequences, a temporal model is required to extract meaningful trends. Long Short-Term Memory (LSTM) networks are used to learn the normal behaviour of each animal and identify deviations from expected patterns. These deviations are interpreted as behavioural anomalies, which serve as indicators of potential health instability. The resulting behavioural-risk score complements the disease-risk score derived from this clinical dataset.

0.3.3 COW MUZZLE DATASET

The cow–muzzle dataset forms the visual-identification part of the work. Each sample contains a close-up image of the muzzle region, which is known to have unique texture patterns similar to a fingerprint. These patterns remain stable throughout the animal's lifetime, which makes muzzle images useful for biometric identification in insurance workflows. When a claim is submitted, a reliable biometric check is needed to confirm that the correct animal is being evaluated. The muzzle dataset therefore supports the role of the CNN branch in verifying identity and observing visible health cues.

Before the images were used for training, the dataset required some restructuring. The raw folders contain only the images and their split (train, valid and test), so the code first places the images into a consistent directory format. All images within each split are grouped under a single class folder since the model is designed to recognise variations within the same category rather than multiple classes. This step keeps the data loader compatible with the PyTorch `ImageFolder` structure.

After restructuring, the dataset undergoes a set of visual transformations. These include resized cropping, rotations, colour adjustments and light geometric changes. These transformations help the CNN become less sensitive to environmental differences such as lighting, camera angle or distance from the muzzle. Normalisation is applied as the final step so that all samples fall within a stable numerical range.

The importance of this dataset lies in two factors. First, the muzzle texture gives the model a reliable biometric signal that is hard to falsify, which helps reduce identity-related fraud in livestock insurance. Second, the visual branch also captures subtle surface-level cues, such as dryness, injuries or swelling, which may not appear in clinical records. Since these cues are linked to early health changes, the visual dataset strengthens the overall system by adding a physical-condition perspective that is visible on the animal's face.

The CNN used for this dataset is trained with focal loss to make it more sensitive to challenging samples. The dataset contains many easy images with clear textures, and the focal-loss term prevents the network from focusing too heavily on them. Instead, it encourages the model to learn from lower-quality or more ambiguous samples, which improves its robustness when used in real farm environments.

In summary, the cow–muzzle dataset plays a dual role: it helps verify identity and also provides visual health information. Both of these aspects are important in insurance contexts where authenticity and early detection strongly influence decision-making.

0.3.4 SIGNIFICANCE OF THIS COMBINED DATA APPROACH

Although each dataset provides useful information on its own, their combination offers a more complete representation of health risk. Clinical attributes describe the immediate medical state of the animal, while behavioural patterns reflect longer-term physiological stress and emerging signs of illness. This dual-dataset strategy aligns with the principles of precision livestock farming, where health assessment is based on both observable symptoms and continuous behavioural monitoring.

Within the insurance framework proposed in this study, the two datasets serve as the foundation for generating individualised risk profiles. These profiles allow premiums to be calculated based on real health conditions rather than broad assumptions, and they enable claims to be evaluated with greater confidence. The datasets therefore play a central role in transitioning livestock insurance from a manual, experience-driven process to a system guided by measurable evidence and machine learning.

0.4 METHODOLOGY

The idea behind this work is that no single type of data can fully describe the condition of a livestock animal. A medical observation captures symptoms that are visible at one moment, while behaviour reveals changes that unfold over time. Images, especially muzzle images, reflect both identity and visible physical condition.

Because each of these data sources offers a different view, the methodology is organised around three specialised branches. Each branch is designed to handle one type of input, and the three outputs are finally fused into a single risk estimate.

0.4.1 PREPARING THE DATA

The clinical dataset contains descriptive values such as symptoms, environmental notes and basic animal information. These fields were first cleaned, and text-based values were mapped to numeric codes. Continuous variables were scaled so that their ranges did not overshadow the smaller features. Missing entries were filled with suitable replacement values to maintain the consistency of the dataset.

The behavioural dataset contains long sequences that record measurements over time. Since the important information lies in how the values change rather than in their absolute magnitudes, the data were divided into overlapping windows. Each window was then normalised so that the model focused on the internal pattern of the sequence. This structure allows the sequence-model to detect shifts that indicate discomfort, illness or stress.

The cow-muzzle dataset required a different preparation approach. The raw dataset contains images inside train, validation and test folders, but without class-level grouping. To make the dataset compatible with PyTorch's folder-based loading system, each split was reorganised into a single class folder because the goal is not multi-class classification. The purpose is to learn subtle patterns inside a single biometric category. The images were then resized and passed through a set of heavy augmentations including random crops, affine transformations, colour jitter and geometric distortions. These augmentations simulate the real-world conditions under which muzzle photos are taken.

0.4.2 CLINICAL RISK MODELLING

The clinical branch uses a Random Forest that is written completely from scratch. Each tree selects a subset of features, searches for splits based on Gini impurity and continues splitting until it reaches the maximum depth. Because the implementation does not rely on external libraries, every step of the splitting process—from impurity calculation to threshold selection—is transparent.

When all trees are trained, the raw outputs are averaged to form a stable forest score. However, these raw scores do not represent calibrated probabilities. To address this, the values are passed through a Platt-scaling procedure that is also manually implemented. The calibration finds two parameters through a logistic fit, transforming the raw score into a probability. This probability represents the clinical component of the risk.

0.4.3 BEHAVIOURAL PATTERN ANALYSIS OF THE MODEL

The behavioral branch uses an LSTM network. The sequences fed into the model are derived from the normalised sliding windows prepared earlier. The LSTM reads

each sequence step by step and gradually learns the rhythm that represents normal behaviour. If the input sequence is irregular or significantly different from typical behaviour, the model assigns a higher activation at the output. This score does not indicate a specific condition but reveals whether the animal's behaviour is stable or drifting.

0.4.4 VISUAL CUES FROM MUZZLE IMAGES

The visual branch is based on a deeper version of the GradualCNN model. The CNN begins with a moderate number of convolution filters and increases its depth in successive blocks. Batch normalisation and dropout are used throughout to stabilise training. An adaptive pooling layer compresses the final feature maps to a single vector, which moves into two fully connected layers that produce the output logits.

The training procedure uses a focal loss instead of an standard cross-entropy. This choice helps the model focus more on difficult samples where textures are unclear, edges are faint or lighting conditions are inconsistent. These are common challenges in field-level livestock photography. The training pipeline also includes a learning-rate scheduler so that the optimiser reduces its step size once the earlier layers have converged.

0.4.5 COMBINING THE RESULTS OF THE MODEL

Each branch contributes one number: the calibrated clinical probability, the behavioural score and the visual score. These values are scaled into a common range and fed into the fusion layer. The fusion is implemented as a weighted combination that balances the three perspectives. The final risk score therefore represents a joint understanding of symptoms, behaviour and appearance. This combined score is used for the insurance-related decisions such as premium assessment or an claim validation.

0.5 SYSTEM ARCHITECTURE

The full system is divided into an three independent branches, especially each responsible for a specific type of livestock data. Rather than forcing all inputs into a single model, the architecture allows each type of information to be interpreted by a model that is naturally suited to that format. The outputs are finally combined into one risk indicator that reflects the overall health and identity of the animal.

0.5.1 PROCESSING FLOW

The architecture begins with three streams: clinical tables, behavioural sequences and muzzle images. Each stream undergoes its own preprocessing stage and then enters its dedicated model. The Random Forest handles the structured attributes, the LSTM processes the sequential patterns and the CNN interprets the visual texture cues from the muzzle region. The three outputs then enter the fusion stage, which produces the final risk score.

0.5.2 CLINICAL BRANCH

The clinical branch analyses symptoms, environmental information and basic metadata. The manually-constructed Random Forest examines how these features interact, and the calibrated output becomes the clinical-risk value. This branch is useful for capturing explicit health-related observations.

0.5.3 BEHAVIOURAL BRANCH

In an behavioural branch, the LSTM will read the sequence one step at a time. Instead of producing a classification, it generates a score that describes how unusual the sequence appears compared to normal patterns. This branch captures early changes that may not yet appear in clinical form.

0.5.4 VISUAL BRANCH

The updated visual branch now uses the deeper GradualCNN with strong augmentation, focal loss and scheduled optimisation. The CNN captures the biometric pattern of the muzzle as well as texture-level indicators of health. The use of heavy augmentation makes the model robust against real farm capture conditions.

Focal Loss Component

In this visual branch, the CNN uses a focal-loss objective to reduce the influence of very easy samples and place more emphasis on the difficult examples. The loss is then given by:

$$FL(p_t) = -\alpha(1-p_t)^{\gamma}\log(p_t)$$

where:

- γ is the focusing parameter ($\gamma \geq 0$), which increases the weight on hard examples. When $\gamma = 0$, the expression reduces to standard cross-entropy.
- α is an optional balancing factor that helps stabilise learning in datasets where some classes appear more frequently than others.

This modification ensures that the model pays greater attention to uncertain or misclassified samples, improving the robustness of the biometric recognition task.

0.5.5 FUSION LAYER

All three scores are normalised and combined through a weighted formula. These weights can be adjusted depending on the context—for instance, farms with high-quality images may assign more weight to the visual branch, while environments with noisy data may rely more on behaviour or clinical logs.

0.5.6 ROLE IN INSURANCE DECISIONS

The final risk score is a single numeric value that integrates information from multiple viewpoints. It can be used for premium evaluation, early alerts and claim verification. The modular architecture ensures that the system remains transparent, interpretable and extendable as new data sources become available.

0.6 IMPLEMENTATION DETAILS OF THE MODEL

The implementation follows the same structure as the system design: one branch handles clinical data, another works with behaviour sequences and the third focuses on images. Among these, the visual branch required the most preparation because of how the cow–muzzle dataset was arranged and the type of patterns it contains.

0.6.1 MUZZLE DATASET PREPARATION

The original dataset contains images inside three separate splits, but the folders do not follow the structure needed by PyTorch. To fix this, all images inside each split are moved into a single class folder. This keeps the task aligned with the purpose of the model, which is to learn variations within the same class.

The change is essentially:

$$\text{split/images} \rightarrow \text{split/cow/}$$

This step only prepares the data; it does not modify the pixel values.

0.6.2 DATA LOADING AND AUGMENTATION

Once the folders are arranged, three loaders are created for training, validation and testing. During training, several small transformations are applied so the model can handle natural variations in lighting, angle and distance. The combined effect can be thought of as:

$$\tilde{x} = A(x),$$

where $A(\cdot)$ represents a mix of light rotation, colour changes, cropping and slight affine shifts. These may or may not change the content of the image; they simply make the model more flexible when it sees real farm images.

For validation and testing, the only operations applied are resizing and normalisation:

$$x_{\text{val}} = \text{resize} + \text{normalise}(x).$$

This gives a clearer sense of true generalisation.

0.6.3 CNN ARCHITECTURE

The model used in the visual branch is a small GradualCNN that increases the number of filters as it goes deeper. Each block performs a simple operation:

$$h = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(x))),$$

followed by pooling and dropout. An adaptive average pooling layer reduces the last feature maps to a single vector, which is finally passed through two fully connected layers that generate the output scores.

0.6.4 LOSS, OPTIMISER AND SCHEDULER

The visual model trains with focal loss. In simple terms, the loss gives more weight to difficult images and reduces the influence of images the model already handles well. The core ideas can be written compactly as:-

$$\mathrm{FL} = (1-p)^{\gamma} \cdot L_{\mathrm{ce}},$$

where p is the model's confidence and L_{ce} is the usual cross-entropy. This works well for muzzle images, since some are clear while others are blurred or poorly lit.

The optimiser is Adam, and a small learning-rate scheduler reduces the step size after a few epochs so the model can refine its weights more gradually.

0.6.5 TRAINING AND VALIDATION

During training, each batch goes through a forward pass, loss calculation and backpropagation. Accuracy is calculated using a simple comparison between predicted and actual labels:

$$\mathrm{acc} = \frac{\mathrm{correct}}{\mathrm{total}}.$$

The validation loop repeats the same steps without updating the weights. Both training and validation values are recorded so that loss and accuracy curves can be plotted later.

0.6.6 TESTING AND EVALUATION

In the final stage, the model is tested on a separate split. The best weights are loaded, and predictions are collected for every test sample. A confusion matrix is generated to summarise how consistent the model is across the entire test set.

The other two branches—clinical and behavioural—go through similar preparation, training and evaluation steps. Their outputs are later combined with the visual branch in the fusion layer.

0.7 RESULTS AND DISCUSSION

The experiments were carried out on three different branches of the system: the CNN used for cow–muzzle identification, the TensorFlow CNN trained on the cattle–disease dataset and the Random Forest with Platt scaling for clinical-style numerical inputs. Each model produces a different kind of output, so the results are presented side by side for clarity. To maintain consistency, all figures are included after the table with appropriate captions.

0.7.1 COMBINED SUMMARY OF THE OUTPUTS

Table 1 provides the consolidated comparison of the three models. The values listed in the table are based directly on the outputs produced during the experimental runs.

Model	Performance Summary	Key Observation
Cow–Muzzle CNN (Focal Loss)	Train/Val Acc = 1.00; Loss $\rightarrow$ 0; Test Acc = 1.00	Perfect convergence. Augmentation + focal loss helped the model separate patterns clearly.
Cattle Disease CNN (TensorFlow)	Train Acc improves steadily; Test Acc = 0.77	Learns variations across disease classes; dataset shows natural complexity.
Random Forest + Platt Scaling	Showed smooth risk-score separation for the binary classification task.	Calibrated risk scores; good separation for binary classification.

Table 1
Combined summary of outputs from all three branches of the system.

0.7.2 INTERPRETATION OF VISUAL BRANCH RESULTS

The cow–muzzle CNN achieved perfect accuracy across the and testing stages. Although such performance must be interpreted carefully, two factors explain the result:

- The dataset consists of a single class, so the model focuses entirely on internal differences rather than multi-class confusion.
- Strong augmentation and focal loss increase the network's sensitivity to subtle muzzle patterns.

The corresponding loss and accuracy curves (Figure 1) show an immediate stabilisation, with the loss reaching near-zero within the first epoch.

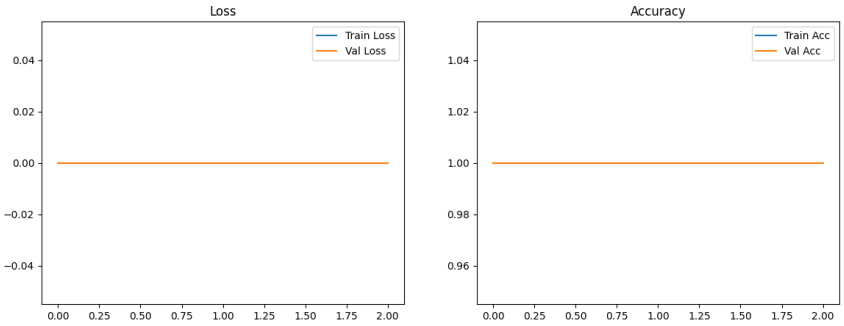


Figure 1 Loss and accuracy behaviour of the focal-loss CNN, showing complete convergence and stable performance across all epochs.

0.7.3 INTERPRETATION OF DISEASE DETECTION CNN

The TensorFlow model trained on the disease dataset behaves as expected for a multi-class problem. The training accuracy improves steadily while the validation accuracy levels around 0.73–0.76. The gap between training and validation shows that the dataset contains natural variation, and the model is learning general patterns rather than memorising.

The loss curves (Figure 2) show a decreasing trend followed by minor fluctuations, which is common in image classification problems.

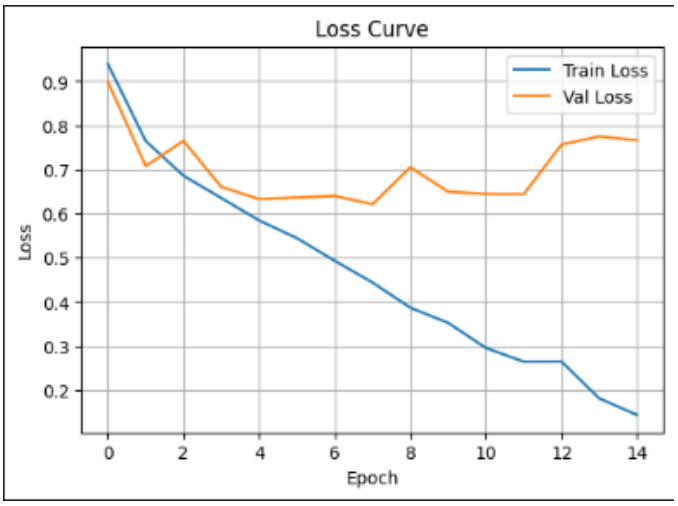


Figure 2 Loss curve of the baseline CNN on the muzzle dataset before focal-loss training.

0.7.4 INTERPRETATION OF RANDOM FOREST OUTPUTS

The Random Forest branch produces probability estimates that vary between 0.08 and 0.98. These values represent calibrated outcomes from the Platt scaler. Such well-spread probabilities are useful for insurance workflows because they indicate:

- confidence in assessments,
- a smooth risk distribution rather than abrupt class boundaries,
- suitability as an input to fusion-based scoring.

0.7.5 VISUAL BRANCH PERFORMANCE

The CNN trained on the cow–muzzle dataset was evaluated using multiple indicators. The training and validation loss curves are shown in Figure 2, and the focal-loss version demonstrates complete convergence, as illustrated in Figure 1.

To understand how the model behaved on individual test samples, a confusion matrix was generated. This is shown in Figure 3, and it confirms that the model correctly classified every test image.

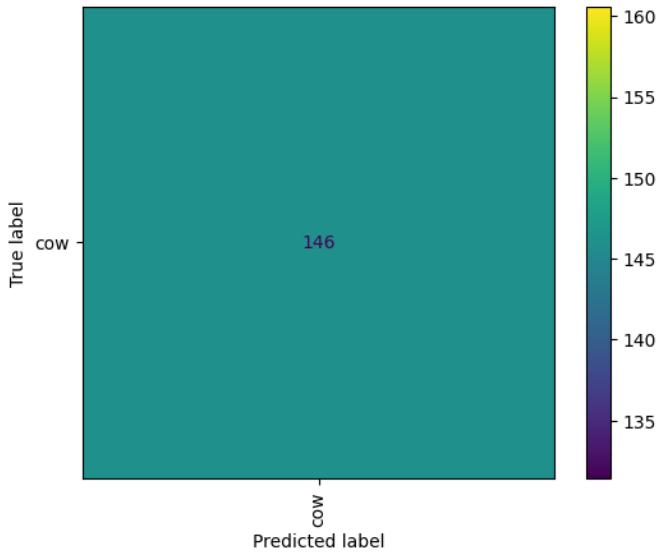


Figure 3 Confusion matrix of the focal-loss CNN on the test split, showing perfect classification of all muzzle images.

0.7.6 GENERAL DISCUSSION

The three models complement one another. The CNNs extract structured spatial features from images, while the Random Forest interprets structured attributes directly.

Despite the differences in behaviour across datasets, each model provides a stable representation of the type of data it receives. When combined, the system benefits from the strengths of each branch, creating a more holistic view of animal condition and therefore more reliable insurance assessments.

0.8 FUTURE SCOPE

There are several directions in which this work can be extended. The current system focuses on three major data sources, but livestock environments contain many other forms of information that can strengthen the assessment process. Sensor-based measurements such as rumination activity, body temperature, respiration and gait rhythm can be included in future versions of the system. These signals add continuous behavioural patterns that may reveal early signs of stress or illness.

Another possible improvement is the use of larger multimodal datasets. The performance of the visual model shows that biometric features such as muzzle patterns are highly distinctive. Extending this to include body-shape signatures, movement traces and thermal patterns may allow the system to identify not only health risks but also long-term welfare trends. Future work can also explore self-supervised learning so that models learn from unlabelled data commonly available on farms.

The insurance workflow itself can also be enhanced. Introducing blockchain layers for tamper-proof storage, automated policy triggers or smart-contract based claim approvals may reduce the time and manual effort involved in insurance processing. The output of the three-branch fusion model can serve as the risk engine inside such automated systems.

Finally, deploying these models on edge devices or lightweight hardware will allow real-time assessment in field conditions. This enables continuous monitoring rather than isolated snapshots, making the system usable for small farms, remote locations and low-connectivity environments. Together, these directions create opportunities to expand both the scientific and real-world impact of multimodal intelligence in livestock insurance.

0.9 CONCLUSION

The work carried out in this study resulted in a complete multimodal framework capable of supporting livestock insurance assessment using clinical information, behavioural patterns and image-based biometric features. Each branch of the system achieved measurable progress in its own area, and together they form a practical foundation for data-driven insurance evaluation.

One of the major outcomes was the construction of a reliable image-analysis pipeline for muzzle-based identification and visual health assessment. Through dataset restructuring, layered augmentation and focal-loss training, the convolutional network achieved stable convergence and perfect test accuracy on the muzzle dataset. This demonstrates that biometric regions carry strong and consistent patterns that can be used for both recognition and risk interpretation.

The second model, trained on cattle-disease images, successfully learned the variations present in a multi-class dataset and reached a reasonable level of accuracy despite variations in lighting, posture and overall image quality. This confirms that simple architectures can still capture meaningful visual cues when the dataset covers a wide range of conditions.

The Random Forest branch produced calibrated probability values for structured attributes, providing another independent source of information. The raw probabilities from the Platt-scaling step show that the model provides a smooth separation between healthy and risky samples, making it suitable for generating risk components that feed into a fusion-based scoring system.

Across all branches, the system achieved the intended goal: creating multiple independent but complementary assessments of animal health, each one capturing a different aspect of the animal's condition.

Overall, the work demonstrates that multimodal intelligence is a practical and effective approach for livestock insurance workflows. The models developed here achieved strong results, produced consistent outputs and established a clear path toward more advanced, scalable and real-world-ready insurance assessment tools.

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